Python Notes

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sir ke notes + NEW = jo add kiya

Step 1: Python kya hai

- Python ek **Scripting Language** hai. Editor: **PyCharm**
- **Interpreted** = code ko line by line chalata hai
- **Interactive** = ek command likho, turant output milta hai
- **Object-oriented** = program ko objects ke form me organize kar sakte hain
  - Class = kisi object ka blueprint / design
  - Object = class se banaya gaya actual item
- **Script Mode** = program ko .py file me save karke poori file run karte hain
- **Bytecode**:
  - hello.py → Python bytecode me convert → hello.pyc

Step 2: print, comment, variable (NEW)

```
print("Hello")        # screen par output
# ye comment hai, Python ignore karta hai
name = "Rinku"        # name = variable
age = 20
```

- **Variable** = data store karne ka dabba
- Naam number se start nahi hota, space nahi hota
  - (my_name sahi hai, my name galat hai)

Step 3: Data Types (NEW)

- **String (str)** = characters ka set " " → "Raj"
- **Integer (int)** = poora number → 25
- **Float (float)** = decimal number → 5.5
- **Boolean (bool)** = True ya False → True

type(25) likhne se type pata chalta hai.

Step 4: input aur Operators (NEW)

```
name = input("Naam likho: ")
age = int(input("Age likho: "))
```

input hamesha string aata hai, isliye int() lagaya.

- **Math:** + - * / % (bacha hissa) ** (power) // (poora bhaag)
- **Compare:** == (barabar) != (barabar nahi) > < >= <=

Step 5: If / Elif / Else

Condition ke True ya False hone ke hisaab se decision leta hai.

- **if** = pehli condition
- **elif** = dusri / teesri condition (multiple check)
- **else** = koi bhi condition True nahi hai

```
marks = 75
if marks >= 80:
    print("A grade")
elif marks >= 60:
    print("B grade")
else:
    print("Fail")
```

Output: B grade

Dhyan rakho: ":" lagana zaroori, andar indentation (4 space) zaroori.

Step 6: While Loop

Jab tak condition True hai, loop chalega.

Condition False = loop ruk jayega.

```
i = 1
while i <= 5:
    print(i)
    i = i + 1
```

Output: 1 2 3 4 5

i = i + 1 bhool gaye to loop kabhi nahi rukega!

Step 7: For Loop

Sequence (list, range, string) ke elements ko ek-ek karke process karta hai.

List:

```
fruits = ["Apple", "Mango", "Banana"]
for fruit in fruits:
    print(fruit)
```

Range:

```
for i in range(1, 6):
    print(i)
```

Output: 1 2 3 4 5

Sudhar: range(1, 6) me 1 se 5 tak aata hai, 6 nahi aata.

Last number hamesha chhoot jata hai.

String:

```
name = "Rinku"
for letter in name:
    print(letter)
```

Step 8: Break aur Continue

- **break** = loop ko poori tarah band (sab band)
- **continue** = current iteration skip, aage chalo

```
for i in range(1, 6):
    if i == 3:
        break
    print(i)
```

Output: 1 2

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

Output: 1 2 4 5

Step 9: List

Ek variable me multiple values store karne ke liye.

Square brackets [ ] me likhte hain. Alag-alag type ka data rakh sakte hain.

```
data = ["Raj", 25, 5.5, True]
fruits = ["Apple", "Mango", "Banana"]
print(fruits[0])     # Apple  (index 0 se shuru)
print(fruits[-1])    # Banana (-1 = last item)
```

List ki value **change ho sakti hai**.

List Functions:

1. **append()** = End me new item add

```
fruits.append("Grapes")
```

2. **insert()** = Specific index par item add

```
fruits.insert(1, "Orange")
```

3. **remove()** = Value ko remove

```
fruits.remove("Mango")
```

4. **sort()** = Ascending order me arrange

```
numbers.sort()
```

5. **reverse()** = Order ulta

```
numbers.reverse()
```

6. **count()** = Value kitni baar aayi

```
numbers.count(10)
```

7. **index()** = Value ka index number

```
fruits.index("Mango")
```

8. **max()** = Sabse bada number
9. **min()** = Sabse chhota number

```
numbers = [10, 20, 50, 80]
print(max(numbers))    # 80
print(min(numbers))    # 10
```

10. **list()** = Sequence ko list me convert

```
name = "Rakesh"
x = list(name)
print(x)    # ['R','a','k','e','s','h']
```

Extra (NEW): len(fruits) = items ki ginti

pop() = last item nikal deta hai

Step 10: Tuple

Ek variable me multiple values store, lekin baad me values **change nahi** kar sakte.

Round brackets ( ) me likhte hain.

```
fruits = ("Apple", "Mango", "Banana")
print(fruits[0])    # Apple
```

fruits = variable, ("Apple","Mango","Banana") = tuple

· **List vs Tuple:** List [ ] badal sakti hai,
Tuple ( ) nahi badal sakta.

Step 11: Dictionary (NEW)

Data **key : value** ki jodi me store hota hai. Curly brackets { } me.

```
student = {"name": "Rinku", "age": 20}
print(student["name"])       # Rinku
student["age"] = 21          # value change
student["city"] = "Indore"   # new item add
```

Step 12: Functions (NEW)

Function = code ka ek block, ek baar likho baar-baar use karo.

· def se function banate hain
· Function **call** karne par hi chalta hai

Simple function:

```
def hello():
    print("Hello Friends")

hello()      # call
```

Parameter ke saath (bahar se value dena):

```
def greet(name):
    print("Hello", name)

greet("Rinku")     # Hello Rinku
```

Return ke saath (result wapas dena):

```
def add(a, b):
    return a + b

result = add(10, 20)
print(result)      # 30
```

Default value:

```
def greet(name="Dost"):
    print("Hello", name)

greet()            # Hello Dost
greet("Raj")       # Hello Raj
```

- **Parameter** = function banate time wala variable (a, b)
- **Argument** = call karte time di gayi value (10, 20)
- **print** sirf dikhata hai, **return** value wapas deta hai

Step 13: Date and Time (NEW)

datetime module use karte hain.

Module = ready-made code ka dabba, import karke use karte hain.

```
import datetime

now = datetime.datetime.now()
print(now)        # 2026-10-07 22:45:30.123456
```

Sirf aaj ki date:

```
today = datetime.date.today()
print(today)      # 2026-10-07
```

Alag-alag part nikalna:

```
print(now.year)      # 2026
print(now.month)     # 10
print(now.day)       # 7
print(now.hour)      # 22
print(now.minute)    # 45
```

```
print(now.second)    # 30
```

Format badalna (strftime):

```
print(now.strftime("%d-%m-%Y"))   # 07-10-2026
print(now.strftime("%H:%M:%S"))   # 22:45:30
print(now.strftime("%d %B %Y"))   # 07 October 2026
```

%d = din %m = mahina number %B = mahine ka naam

%Y = saal %H = ghanta (24 hr) %M = minute

%S = second %A = din ka naam (Wednesday)

Time ruk-ruk ke chalana:

```
import time
print("Start")
time.sleep(2)     # 2 second ruko
print("End")
```

Quick Yaad Rakhne Wali Baatein

- ":" aur indentation har if / loop / function me zaroori
- Index hamesha **0** se shuru hota hai
- range(1, 6) ka last number nahi aata
- List [ ], Tuple ( ), Dictionary { }
- Function ko call karna mat bhoolo

~ All the best! ~